

LINUX FUNDAMENTALS – WEEK 1 LECTURE NOTES

Introduction to Linux

Linux is an open-source, Unix-like operating system developed by Linus Torvalds. It is widely used in servers, cloud computing, embedded systems, cybersecurity, and software development.

Features of Linux

- Open Source
- Multiuser
- Multitasking
- Secure
- Portable
- Stable
- Command Line Support
- GUI Support

Learning Objectives

- Understand Linux architecture.
- Navigate the Linux filesystem.
- Use command-line tools effectively.
- Manage files and directories.
- Configure Linux systems.
- Install and manage software packages.
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Linux Philosophy and Concepts

Core Concepts

Open Source Software

The source code is publicly available and can be modified or distributed by anyone.

Linux Architecture

Hardware → Kernel → Shell → User Applications

Kernel

The kernel is the core component of Linux responsible for:

- Process Management
- Memory Management
- Device Management
- File System Management

Shell

The shell acts as an interface between the user and the kernel.

Examples:

- Bash
- Zsh
- Fish

Linux Principles

- Everything is a File
- Small Modular Tools
- Command-Line Efficiency
- User Permissions and Security
- Flexibility and Portability

Linux Basics and System Startup

Linux Boot Process

1. Power On
2. BIOS/UEFI Initialization
3. Bootloader (GRUB)
4. Kernel Loading
5. System Initialization
6. Service Startup
7. User Login

Important Terms

BIOS

Basic Input Output System used to initialize hardware during startup.

UEFI

Modern replacement for BIOS with enhanced features.

Bootloader

Loads the Linux kernel into memory.

Example: GRUB (Grand Unified Bootloader)

Daemon

Background process running continuously in Linux.

Examples:

- sshd
- cron
- systemd

Startup Commands

Check system information:

```
uname -a
```

Check kernel version:

```
uname -r
```

Shutdown system:

```
sudo shutdown now
```

Restart system:

```
sudo reboot
```

Graphical Interface

Desktop Environment

A Desktop Environment provides graphical interaction with the system.

Popular Desktop Environments:

- GNOME

- KDE Plasma
- XFCE
- Cinnamon

Features

- Desktop
- File Manager
- Panels
- Workspaces
- Application Launcher
- Settings Manager

Desktop Operations

- File Browsing
- Window Management
- Display Configuration
- Theme Customization
- Workspace Management

System Configuration Using GUI

User Management

Add User:

```
sudo adduser username
```

Delete User:

```
sudo deluser username
```

Network Configuration

View IP Address:

```
ip a
```

Check Hostname:

```
hostname
```

Change Hostname:

```
sudo hostnamectl set-hostname newname
```

System Monitoring

View Running Processes:

`top`

View Memory Usage:

`free -h`

View Disk Usage:

`df -h`

Common Linux Applications

Web Browsers

- Firefox
- Chromium

Office Applications

- LibreOffice

Media Players

- VLC Media Player

Text Editors

- Nano
- Vim
- Gedit

Development Tools

- VS Code
- Eclipse
- IntelliJ IDEA

Compression Utilities

- zip
- unzip
- tar

Command Line Operations

Terminal Basics

Open Terminal:

Ctrl + Alt + T

Current User:

`whoami`

Current Directory:

`pwd`

Display Date:

`date`

Display Calendar:

`cal`

Clear Terminal:

`clear`

Directory Navigation

List Files and Directories:

`ls`

Detailed Listing:

`ls -l`

Show Hidden Files:

`ls -la`

Change Directory:

`cd directory_name`

Move to Home Directory:

`cd ~`

Move Back One Directory:

```
cd ..
```

Move to Root Directory:

```
cd /
```

Print Working Directory:

```
pwd
```

File and Directory Management

Create File

```
touch file.txt
```

Create Directory

```
mkdir myfolder
```

Remove Empty Directory

```
rmdir myfolder
```

Remove File

```
rm file.txt
```

Remove Directory Recursively

```
rm -r myfolder
```

Copy File

```
cp source.txt destination.txt
```

Move/Rename File

```
mv oldname.txt newname.txt
```

Viewing File Contents

Display Entire File:

```
cat file.txt
```

View File Page by Page:

```
less file.txt
```

View First 10 Lines:

```
head file.txt
```

View Last 10 Lines:

```
tail file.txt
```

Hard Links and Symbolic Links

Hard Link

```
ln file1 hardlink
```

Symbolic Link

```
ln -s file1 symlink
```

List Inode Numbers:

```
ls -li
```

Searching for Files

Locate Command

Search File:

```
locate filename
```

Update Database:

```
sudo updatedb
```

Find Command

Find File by Name:

```
find /home -name file.txt
```

Find Directories:

```
find /home -type d
```


Find Large Files:

```
find / -size +10M
```

Find Files Modified in Last Day:

```
find /home -mtime -1
```

Wildcards

All Text Files:

```
ls *.txt
```

Files Starting with “a”:

```
ls a*
```

Single Character Match:

```
ls file?.txt
```

Input and Output Redirection

Redirect Output:

```
ls > output.txt
```

Append Output:

```
ls >> output.txt
```

Redirect Input:

```
cat < input.txt
```

Redirect Errors:

```
command 2> error.txt
```

Pipes

A pipe passes output from one command as input to another.

Example:

```
ls -l | less
```

Count Files:

```
ls | wc -l
```

Search Text:

```
cat file.txt | grep Linux
```

User Privileges and Sudo

Execute Command as Administrator:

```
sudo command
```

Switch User:

```
su username
```

Switch to Root:

```
sudo -i
```

Package Management

APT Package Manager (Ubuntu/Debian)

Update Package List:

```
sudo apt update
```

Upgrade Installed Packages:

```
sudo apt upgrade
```

Install Package:

```
sudo apt install package_name
```

Remove Package:

```
sudo apt remove package_name
```

Search Package:

```
apt search package_name
```

List Installed Packages:

```
apt list --installed
```

Lab Exercises Completed

Lab 7.1

Killing the Graphical User Interface

Lab 7.2

Locating Applications

Lab 7.3

Creating, Moving and Removing Files

Lab 7.4

Finding Directories and Creating Symbolic Links

Lab 7.5

Installing and Removing Software Packages

Key Takeaways

- Learned Linux architecture and operating principles.
- Understood Linux boot process and startup sequence.
- Explored graphical desktop environments.
- Configured Linux systems using GUI tools.
- Practiced command-line operations.
- Managed files and directories efficiently.
- Used links, wildcards, pipes, and redirection.
- Installed and removed software packages.
- Gained hands-on experience through laboratory exercises.
- Created a GitHub repository to document internship activities and learning progress.